

India emerges as a country with boundless potential in this field. The remarkable progress observed in India's automotive and manufacturing sectors reveals an exciting pathway for the evolution of vehicular displays. While many across the globe applaud the achievements of companies like Tata Motors in the expansive automotive industry, a wealth of opportunities exists in the display segment, awaiting India's exploration. Given the progress in LCD technology and the anticipated transition to micro-LEDs, India's burgeoning manufacturing infrastructure is well-suited to lead this revolution. Recognising this potential, our study incorporates India's prospective contribution to the industry's trajectory. India is not only a hub of manufacturing expertise but also on the cusp of pioneering breakthroughs, notably in domains like micro-LEDs. The horizon looks promising, with India poised to become a cornerstone in the evolution of automotive displays.

A recent conversation with Dr Miguel El Guendy, Technology Analyst at IDTechEx, provided profound insights into this dynamic landscape.

Key automotive display technologies

The realm of automotive displays is extensive and continually evolving, with two technologies, LCD and OLED, emerging as leaders in this domain. These technologies reign supreme in the automotive space and also have a significant presence in the broader display industry.

LCD dominance in vehicle displays. Liquid crystal display (LCD) presently stands as the most prevalent technology for automotive displays. Its dominance is rooted in several factors, with cost-effectiveness being the primary driver. LCD strikes a commendable balance

How does the rise of EVs impact the automotive display market?

The surge in electric vehicles (EVs) significantly influences the automotive display market. While displays are becoming a standard feature across all types of vehicles, from HUDs to side window displays and dashboard screens, there is a shared emphasis on reducing power consumption. The importance of energy efficiency extends beyond EVs and has gained increasing significance due to the rising adoption of EVs. Saving energy and promoting sustainability have become key trends in automotive display design. Manufacturers strive to cater to a diverse market, spanning premium vehicles and more budget-friendly options. For lower-tier vehicles with tighter margins, cost-effective choices like LCDs are preferred, whereas premium vehicles might opt for technologies like OLEDs. Regardless of the type of vehicle, the common thread remains the focus on minimising power consumption, aligning with the broader shift toward sustainable mobility solutions.

between image quality and brightness, making it a suitable choice for many vehicles. LCD technology has matured, ensuring reliability and economic viability.

The vast supplier network for LCDs further solidifies its commanding position. Prominent names in the LCD market include Tianma, BOE, and AUO. These suppliers ensure competitive pricing and a wide range of features to cater to diverse automotive needs. Additionally, several others are ready to fill the gap if one supplier needs to meet a manufacturer's specific requirements. This flexibility and availability make LCD the preferred choice for many automakers, especially those producing vehicles with tighter profit margins.

Elevating premium car displays with OLEDs. Organic light emitting diodes (OLEDs) are another significant player in the automotive display sector, although they currently occupy a niche space. OLEDs are typically found in premium vehicles, primarily due to their higher cost. However, OLED's superior contrast ratio sets it apart, arguably providing better image quality than LCDs.

Industry titans such as Samsung and LG Display take the lead in the OLED landscape. Both are renowned for their collaborations with premium vehicle manufacturers, emphasising the luxurious user experience

that OLEDs can deliver. The richness of colours and the depth of blacks in OLED displays make them a preferred choice for high-end vehicles. Nevertheless, the cost factor acts as a deterrent for mass-market cars, where margins are slimmer, and the budget for components like displays is constrained.

A display revolution in automotive technology

The dynamic world of automotive displays is on the brink of another transformation. While LCDs and OLEDs have dominated the industry, a fresh player, microLEDs, is now entering the arena. These next-generation displays are poised to bring numerous benefits, offering superior performance compared to their current counterparts.

MicroLEDs enhance attributes like greater brightness and overall performance in comparison to LCDs and OLEDs. However, they are still in the early stages of commercial application. The primary challenges to their widespread adoption are their high cost and less-than-optimal yields. Given these challenges, it is unlikely we will witness a significant penetration of microLEDs into the market in the near future, possibly within the next few years.

Nevertheless, there is palpable excitement about what microLEDs can bring to the table. Their ad-